

Phase q-window and Delta-refinement audit

Status: input points and HPC helper scripts prepared; exact reruns are pending.

Input Counts

```
{  
  "qwindow_sensitive_points": 342,  
  "delta_sensitive_points": 96,  
  "clean_control_points": 20,  
  "combined_unique_points": 345  
}
```

Scope

This is an audit-only production-oriented numerical robustness update. It does not modify acquisition, StopController, NN training, or the active-learning dataset.

Required interpretation rules

- q-window expansion is not required to prove superconductivity once a lower-free-energy positive-Delta superconducting state is already found.
- q-window expansion is required to check branch identity, q-opt stability, boundary robustness, and future topology classification.
- eta response has already been downgraded to response-extraction pathology unless 'eta_response_valid=True'.